

HENG QUAN (权衡)

Princeton University, Atmospheric and Oceanic Sciences Program, US

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EDUCATION & WORK EXPERIENCE

Princeton University

Sep. 2022 – June. 2027 (expected)

PhD in Atmospheric and Oceanic Sciences Program (Advisor: Stephan Fueglistaler)

Woods Hole Oceanographic Institution

June. 2025 – Aug. 2025

Geophysical Fluid Dynamics Fellow

Peking University

Sep. 2018 - Jul. 2022

Bachelor of Atmospheric Sciences (School of Physics)

PUBLICATIONS

Submitted:

1. **Quan, H.**, Yang, D., Boos, W., Shaw, T., Ge, H., Zeng, Y., & KleinStern, C. Vapor buoyancy and the African easterly jet. (under review)

Refereed publications:

1. **Quan, H.**, Zhang, Y., Dagan, G., & Fueglistaler, S. (2026). Periodic extreme rainfall in a warmer climate due to stronger convectively-coupled waves. *Science Advances*, 12(32).
2. **Quan, H.**, Zhang, Y., Dagan, G., & Fueglistaler, S. (2026). Transition to double-cell mock Walker circulations with surface warming explained by periodic convection. *Geophysical Research Letters*, 53(18).
3. **Quan, H.**, Zhang, Y., & Fueglistaler, S. (2025). Weakening of tropical free tropospheric temperature gradients with global warming. *Journal of the Atmospheric Sciences*, 82(1): 31-43.
4. **Quan, H.**, Fueglistaler, S., Zhang, B., & Wang, C. (2025). The Sea Surface Temperature Pattern Effect on Outgoing Longwave Radiation: The Role of Large-scale Convective Aggregation. *Geophysical Research Letters*, 52(11).
5. **Quan, H.**, Koll, D., Lutsko, N., & Yuval, J. (2025). Optimal Solar Geoengineering Strategies Based on Reinforcement Learning. *Journal of Geophysical Research: Atmospheres*, 130(23).
6. **Quan, H.**, Fueglistaler, S., Zhang, B., & Wang, C. (2024). Non-linear radiative response to patterned global warming due to convection aggregation and non-linear tropical dynamics. *Journal of Climate*, 37(21): 5675-5686.
7. **Quan, H.**, Zhang, B., Bourguet, S., Linz, M., & Chen, G. (2023). How Do Different Processes Shape Temperature Probability Distributions? A Percentile-averaged Temperature Tendency Decomposition. *Journal of Climate*, 36(15): 5179-5196.
8. **Quan, H.**, Chai, W., & Fu, Z. (2022). Asymmetry of daily mean temperature series over China and its frontal mechanism. *International Journal of Climatology*, 42(3), 1828-1840.
9. Chai, W., Huang, Y., Yang, L., **Quan, H.**, & Fu, Z. (2021). Evaluation of re-analyses over China based on the temporal asymmetry of daily temperature variability. *Theoretical and Applied Climatology*, 1-13.

INVITED TALKS

- Departmental seminar, Department of Atmospheric and Oceanic Sciences, Peking University, Nov 2026
- Climate, Atmosphere, and Ocean Seminar, The Hebrew University of Jerusalem, June 2026
- Departmental seminar, Department of Atmospheric and Oceanic Sciences, Peking University, Oct 2024
- Departmental seminar, NYU Courant Institute, Center for Atmosphere Ocean Science, Sep 2024

CONFERENCE TALKS

- 25th Atmospheric and Oceanic Fluid Dynamics Conference, Houston, TX, Jan 2026
- AGU Fall Meeting 2025, New Orleans, LA, Dec 2025 (1)
- AGU Fall Meeting 2025, New Orleans, LA, Dec 2025 (2)
- AGU Fall Meeting 2024, Washington D.C., Dec 2024
- ICDM Workshop 2024, Nanjing, Oct 2024
- 24th Atmospheric and Oceanic Fluid Dynamics Conference, Burlington, VT, June 2024
- CFMIP Meeting 2024, Boston, MA, June 2024
- AGU Fall Meeting 2023, San Francisco, CA, Dec 2023

POSTERS

- CFMIP Meeting 2026, Taipei, Oct 2026
- 25th Atmospheric and Oceanic Fluid Dynamics Conference, Houston, TX, Jan 2026
- AGU Fall Meeting 2024, Washington D.C., Dec 2024
- Gordon Research Conference on Radiation and Climate, Leviston, Maine, July 2023
- CFMIP Meeting 2023, Paris, July 2023

AWARDS & HONORS

PhD (2022-now):

- Charlotte Elizabeth Procter Fellowship, Princeton University, 2026-2027
- Geophysical Fluid Dynamics Fellow, Woods Hole Oceanographic Institution, 2025
- Best Student Oral Presentation Award in the 2024 ICDM workshop
- Accepted by the 2024 Rossbypalooza summer school at University of Chicago

Undergraduate (2018-2022):

- National Scholarship (**top 1%** among 176 students, 2021)
- Merit Student Pacesetter (2021)
- POSCO Asia Scholarship (**top 3%** among ~180 students, **awarded three times** in 2019, 2020 and 2021)
- Outstanding bachelor student in Beijing (**10** among 176 students, 2022)
- Weiming bachelor student (**7** among 176 students, 2022)

SERVICE

- Session convener in AGU 2026 meeting
- Organizer of the 2026-2027 weekly student & postdoc seminar at Princeton University, AOS program
- Organizer of the 2025-2026 “climate chedann” weekly climate dynamics seminar among ~500 scholars
- Teaching assistant, GEO 361 “Earth’s Atmosphere”, fall 2023, Princeton University